Class 12 · Physics

Chapter 4: Motion in a Plane

Worksheet — Motion in a Plane

Total marks: **7** · Questions: **3**

Instructions: Attempt all questions. Take $g = 10\ m/s^2$ unless otherwise stated. Show steps for numerical questions.

Q1. On level ground, the range of a projectile is maximum at a launch angle of [1 mark]

- (A) 30°
- (B) 45°
- (C) 60°
- (D) 90°

Q2. A projectile is launched at 20 m/s at 30°. Find T, H, R. ($g = 10\ m/s^2$) [3 marks]

Q3. Derive expressions for the time of flight, maximum height and horizontal range of a projectile. [3 marks]

Answer Key

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Q1. On level ground, the range of a projectile is maximum at a launch angle of

Answer: 45°

$R = u^2 \sin 2\theta / g$; $\sin 2\theta$ is maximum at $2\theta = 90°$.

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Q2. A projectile is launched at 20 m/s at 30°. Find T, H, R. ($g = 10$ m/s^2)

Answer: $T = 2 \times 20 \times 0.5 / 10 = 2$ s. $H = 400 \times 0.25 / 20 = 5$ m. $R = 400 \times \sin 60° / 10 = 20\sqrt{3} \approx 34.6$ m.

Apply $T = 2u \sin\theta / g$, $H = u^2 \sin^2\theta / 2g$, $R = u^2 \sin 2\theta / g$.

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Q3. Derive expressions for the time of flight, maximum height and horizontal range of a projectile.

Answer: Take launch as origin, x horizontal, y up. $u_x = u \cos\theta$, $u_y = u \sin\theta$. $a_x = 0$, $a_y = -g$. Then $x(t) = u \cos\theta \cdot t$ and $y(t) = u \sin\theta \cdot t - \frac{1}{2} g t^2$. Setting $y = 0$ gives $T = 2u \sin\theta / g$. At apex $v_y = 0$ $\Rightarrow H = u^2\sin^2\theta / (2g)$. $R = u_x \times T = u^2 \sin 2\theta / g$.

Apply 1D kinematics component-wise.

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